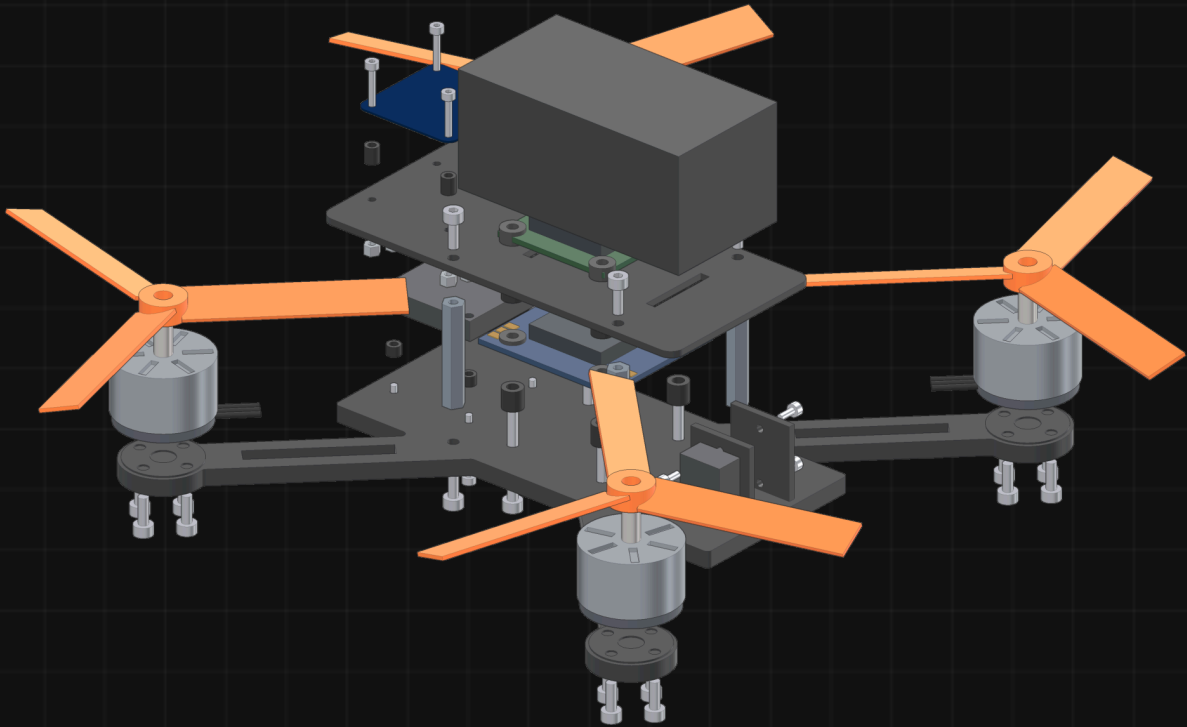


ASSEMBLY GUIDE

5-INCH 6S QUADCOPTER

Reference 5-inch 6S freestyle quadcopter: DolphinRC F405 V3 + AM32 60A stack, four MEPS NEON 2207 V2 1950KV motors with HQProp Ethix S5 props, CNHL 1100 mAh 6S pack, RadioMaster RP1 V2 ELRS receiver, DJI O4 Air Unit, Matek M9N-5883 GNSS/compass, custom frame.



Exploded along each joint's axis in fastener-stack order (14 mates, 72 hardware parts, battery by placement hint).

STEPS	MATES	HARDWARE	PARTS
11	14	72	93
mechanical and wiring	from assemble()	screws, spacers, standoffs, nuts	27 BOM lines

■ HOW THIS GUIDE IS MADE

Steps, parts and hardware come from the UHD model: mates and fastener stacks from `assemble()`, wiring from `wiringChecklist`. The coverage table at the end lists where every mate, fastener and wire appears.

- A Assumption: the model marks this value as an assumption (assumption trait)
- D Derived: computed from model values; formula in the provenance appendix
- Unmarked values are cited from the part's sources (`sources.json`) or are design values of the model.

01 BEFORE YOU BEGIN

CONTENTS

01	Before you begin	02
02	Kit contents	03
03	Hardware	04
04	Mount the motors	06
05	Build the FC and ESC stack	07
06	Solder the motor leads	08
07	Solder the battery lead and capacitor	09
08	Install the air unit and camera	10
09	Connect the air unit	11
10	Connect the receiver	11
11	Fit the top plate	12
12	Mount and connect the GNSS	13
13	Strap on the battery	14
14	Fit the propellers	15
15	Coverage	16

TOOLS

- **1.5 mm hex key** for M2 x 18 mm SHCS, M2 x 5 mm camera screw, M2 x 12 mm SHCS (ISO 4762 socket size)
- **2.5 mm hex key** for M3 x 30 mm SHCS, M3 x 8 mm SHCS, M3 x 8 mm motor screw
- **4 mm nut driver** for M2 nyloc nut, DIN 985, A2 stainless
- **5.5 mm nut driver** for M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless
- **5.5 mm spanner** to hold M3 aluminium standoff, 30 mm, female-female, 5.5 mm AF
- **Soldering iron, solder, heat-shrink**

Derived from the fasteners in the model: hex keys from each screw's stated socket or the ISO 4762 socket size for its thread, drivers and spanners from each nut's and standoff's across-flats.

READING THE FIGURES

-  Already assembled
-  Fitted in this step
-  Hardware for this step
-  Assembly axis

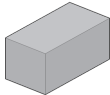
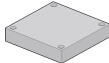
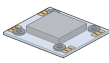
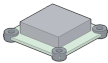
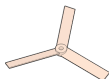
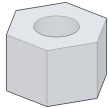
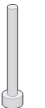
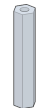
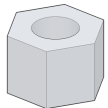
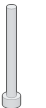
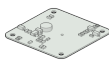
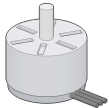
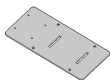
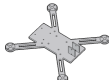
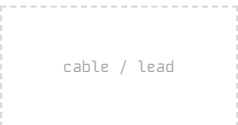
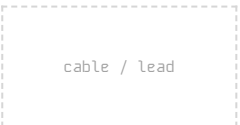
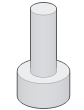
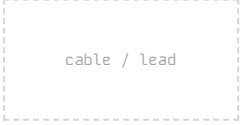
Parts are drawn exploded along each joint's axis in stacking order, from the model's fastener stacks and mate gaps.

WARNING Remove the propellers for any work with the battery connected. Fit them last (step 11).

NOTE Each step's torque line gives the tightening torque of the part turned, from the harness's fastener stack. 8 of them are **assumptions** (marked A): no frame, motor or DJI source states a torque, so they are derated from supplier tables for the screw grade into steel. Threadlocker is only advised where a harness says so (steel into aluminium).

02 KIT CONTENTS

GENERATED/BOM.JSON

 ×1 CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60) cnhl-black-series-1100mah-6s-100c	 ×1 DJI O4 Air Unit dji-o4-air-unit	 ×1 DolphinRC AM32 60A 4-in-1 ESC dolphinrc-am32-60a-4in1-es- c	 ×1 DolphinRC F405 V3 Flight Controller dolphinrc-f405-v3-flight- controller	 ×8 ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 ettinger-005-83-060-m3-ny- lon-spacer-6mm
 ×1 HGLRC Kevlar battery strap 20x250mm hglrc-kevlar-battery-stra- p-20x250mm	 ×4 HQProp Ethix S5 5x4x3 Propeller hqprop-ethix-s5	 ×8 M2 nyloc nut, DIN 985, A2 stainless iso-10511-m2-nyloc-nut	 ×4 M2 nylon spacer, Ø2.7/4.0 x 3 mm m2-nylon-spacer-3mm	 ×4 M2 nylon spacer, Ø2.7/4.0 x 5 mm m2-nylon-spacer-5mm
 ×4 M2 x 12 mm SHCS iso-4762-m2x12	 ×4 M2 x 18 mm SHCS iso-4762-m2x18	 ×4 M3 aluminium standoff, 30 mm, female-female, 5.5 mm AF m3-aluminium-standoff-30m- m	 ×4 M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless iso-10511-m3-nyloc-nut	 ×4 M3 x 30 mm SHCS iso-4762-m3x30-socket-hea- d-cap-screw
 ×8 M3 x 8 mm SHCS iso-4762-m3x8	 ×1 Matek M9N-5883 GNSS & Compass matek-m9n-5883	 ×4 MEPS NEON 2207 V2 1950KV Brushless Motor meps-neon-2207-v2-1950kv	 ×1 Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor panasonic-eeu-fr1v471	 ×1 RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver radiomaster-rp1-v2-elrs-2- g4
 ×1 5-inch frame top plate quadcopter-5in-top-plate	 ×1 5-inch quadcopter frame quadcopter-5in-frame	 ×1 DJI O4 3-in-1 cable dji-o4-3in1-cable	 ×1 FC/ESC SH1.0 8-pin cable dolphinrc-sh8-fc-esc-cabl- e	 ×4 M2 x 5 mm camera screw o4-camera-screw-m2x5
 ×16 M3 x 8 mm motor screw motor-screw-m3x8	 ×1 XT60 battery lead (12 AWG) dolphinrc-xt60-battery-le- ad			

Quantities from the generated BOM; drawings are the parts' CAD artifacts, not to scale.

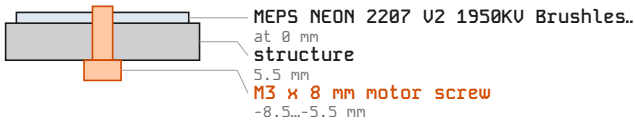
03 HARDWARE

FASTENER STACKS

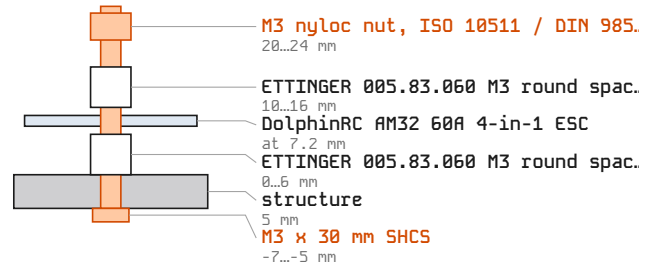
Every screw, spacer, standoff and nut is placed by a harness's `fastenerStack`.
The sections below are drawn from those positions and the mate gaps: one hole of each joint, to one scale, millimetres from the structure face.

HARNESS	PARTS PLACED	TORQUE	USED IN
Motor hardware (M3) ×4 joint arm_fr_mount, arm_rr_mount, arm_rl_mount, arm_fl_mount	16 × M3 × 8 mm motor screw	0.6 N·m ^A M3 × 8 mm motor screw	Step 1
Stack hardware (M3) joint stack_mount	4 × M3 × 30 mm SHCS 8 × ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 4 × M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	0.3 N·m ^A M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	Step 2
Frame hardware (M3 standoffs) joint top_plate_mount	8 × M3 × 8 mm SHCS 4 × M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF	0.6 N·m ^A M3 × 8 mm SHCS 0.6 N·m ^A M3 × 8 mm SHCS	Step 8
Camera hardware (M2) joint camera_mount	4 × M2 × 5 mm camera screw	0.1 N·m ^A M2 × 5 mm camera screw 0.1 N·m ^A M2 × 5 mm camera screw	Step 5
Air unit hardware (M2) joint vtx_mount	4 × M2 × 18 mm SHCS 4 × M2 nylon spacer, Ø2.7/4.0 × 3 mm 4 × M2 nyloc nut, DIN 985, A2 stainless	0.1 N·m ^A M2 nyloc nut, DIN 985, A2 stainless	Step 5
GPS hardware (M2) joint gps_mount	4 × M2 nyloc nut, DIN 985, A2 stainless 4 × M2 nylon spacer, Ø2.7/4.0 × 5 mm 4 × M2 × 12 mm SHCS	0.1 N·m ^A M2 × 12 mm SHCS	Step 9

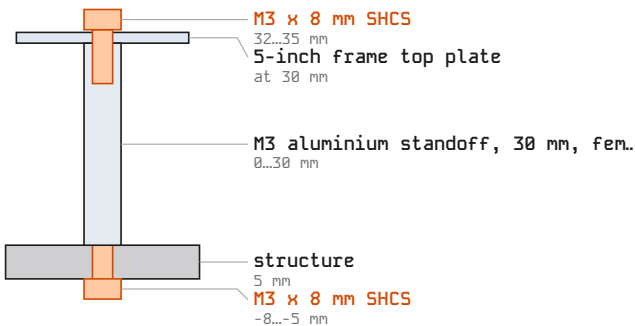
Motor hardware (M3) quadcopter-5in-motor-hardware



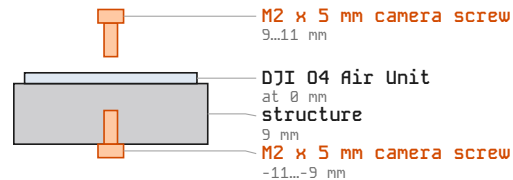
Stack hardware (M3) quadcopter-5in-stack-hardware



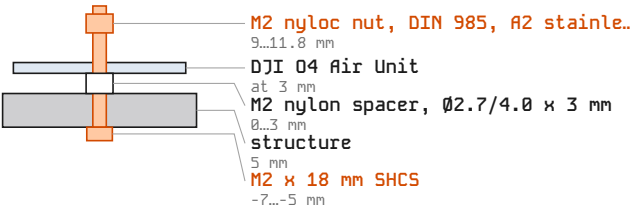
Frame hardware (M3 standoffs) quadcopter-5in-frame-hardware



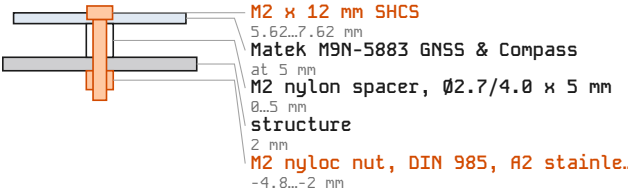
Camera hardware (M2) quadcopter-5in-camera-hardware



Air unit hardware (M2) quadcopter-5in-vtx-hardware



GPS hardware (M2) quadcopter-5in-gps-hardware



1 MOUNT THE MOTORS

mates: arm_fr_mount, arm_rr_mount, arm_rl_mount,
arm_fl_mount

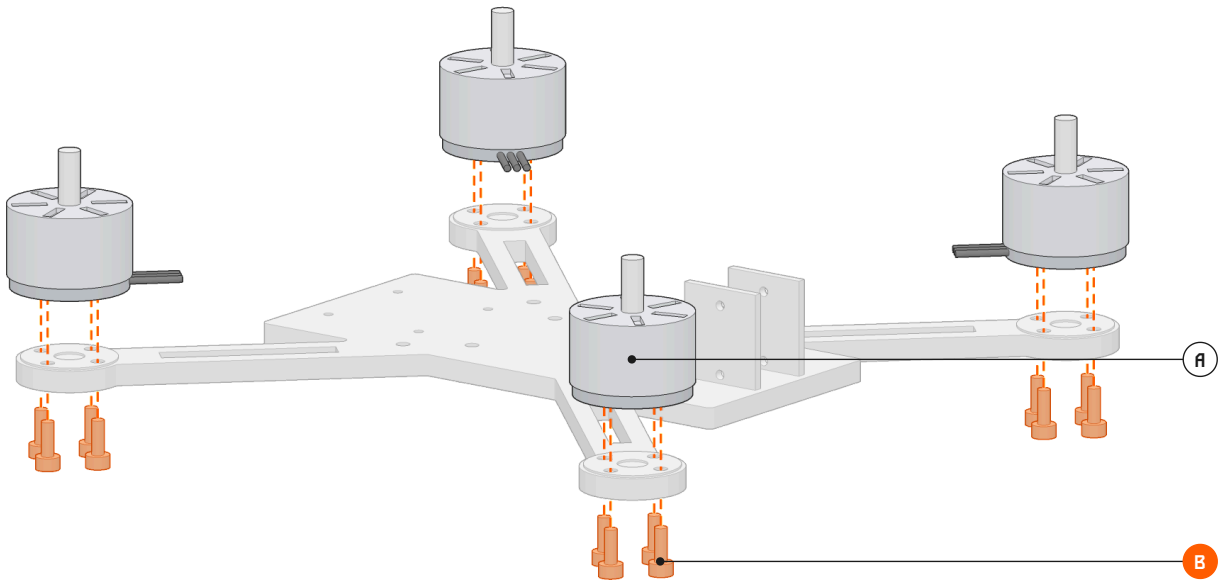
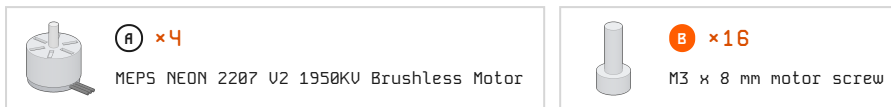


Figure 1 Mount the motors: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Insert 4 × M3 x 8 mm motor screw from below, up through the motor mount.
2. Seat a MEPS NEON 2207 V2 1950KV Brushless Motor onto each motor mount (turned 180° / 270° / 90° by corner, per the mates). Repeat on all 4.

Torque 0.6 N·m **A** M3 x 8 mm motor screw Tools 2.5 mm hex key Step 1 of 11

TIP Point each motor's leads back along its arm toward the stack; the rotation on the pad is set by the mate.

TIP Threadlocker: medium strength (blue) on the motor screws. Steel into aluminium with no locking element, so vibration can back the screws out. Advice is from X-Team's motor-screw guide, a secondary source; no maker of these parts states it. – quadcopter-5in-motor-hardware

2 BUILD THE FC AND ESC STACK

notes: stack_mount, fc_mount · links: stack/fc_esc_cable


A ×1
DolphinRC AM32 60A 4-in-1 ESC


B ×1
DolphinRC F405 V3 Flight Controller


C ×4
M3 x 30 mm SHCS


D ×8
ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25


E ×4
M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless


F ×1
FC/ESC SH1.0 8-pin cable

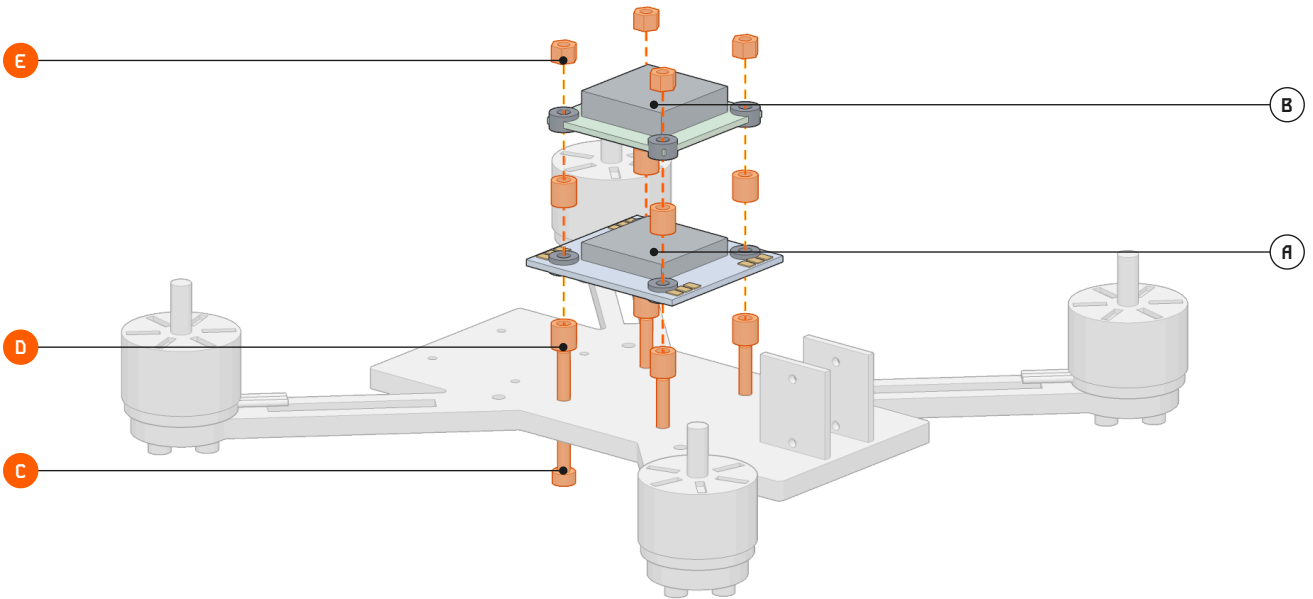


Figure 2 Build the FC and ESC stack: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Insert 4 × M3 x 30 mm SHCS from below, up through the stack mount.
2. Slide 4 × ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 over the screws.
3. Lower the DolphinRC AM32 60A 4-in-1 ESC onto the stack mount (7.2 mm gap).
4. Slide 4 × ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 over the screws.
5. Lower the DolphinRC F405 V3 Flight Controller onto the stack mount (17.2 mm gap).
6. Finish with 4 × M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless; tighten until snug.
7. Using the FC/ESC SH1.0 8-pin cable, connect FC/ESC 8-pin cable as in the table (7 connections).

LINK	FROM	TO	TERMINATION	DRC
FC/ESC 8-pin cable	Flight controller M1 · M2 · M3 · M4 · VBAT · GND · CUR	4-in-1 ESC M1 · M2 · M3 · M4 · V · G · CUR	jst_sh_8 / jst_sh_8	CONFIGURED

Torque 0.3 N·m ^A M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless Tools 2.5 mm hex key, 5.5 mm nut driver Step 2 of 11

i **NOTE** Per post, bottom to top: screw head under the 5 mm bottom plate, 6 mm spacer, ESC in its grommet (4 mm), 6 mm spacer, FC in its grommet (4 mm), nyloc nut; 1 mm of thread protrudes. Boards use the M3 x 8 grommets included with the DolphinRC stack. Stack-up worked out in the fastener parts' usage notes (assumes a 5 mm plate and 4 mm grommet clamp). — quadcopter-5in-stack-hardware

NOTE Included with the DolphinRC F405 V3 stack (manual: 'SH1.0mm 8Pin Cable 50mm'). Straight-through, so pin 1 maps to pin 1. – dolphinrc-sh8-fc-esc-cable

3

SOLDER THE MOTOR LEADS

links: m2_phases, m1_phases, m3_phases, m4_phases

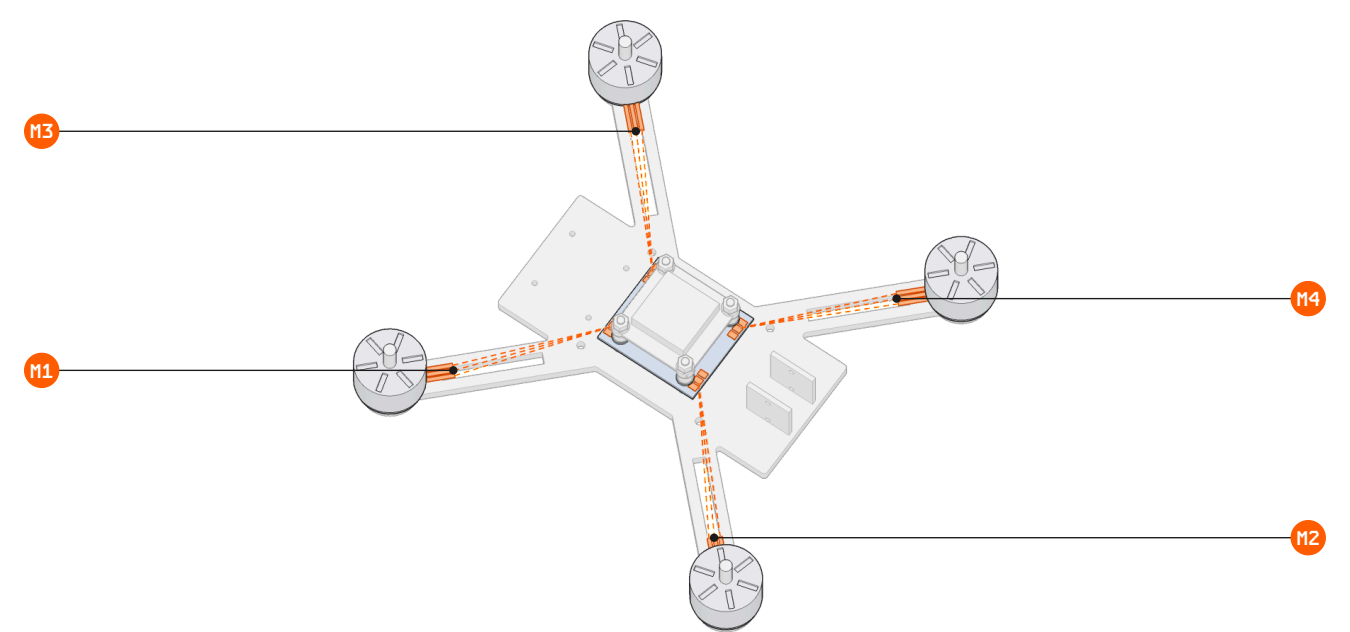


Figure 3 Each motor's three leads run to its ESC channel pads (accent); routes from the interfaces' geometry frames.

1. Connect Motor 2, Motor 1, Motor 3, Motor 4 as in the table (12 connections).

LINK	FROM	TO	TERMINATION	DRC
Motor 2	4-in-1 ESC M2 PHASE_A · M2 PHASE_B · M2 PHASE_C	Motor A · B · C	bare_wire_lead	CONFIGURED
Motor 1	4-in-1 ESC M1 PHASE_A · M1 PHASE_B · M1 PHASE_C	Motor A · B · C	bare_wire_lead	CONFIGURED
Motor 3	4-in-1 ESC M3 PHASE_A · M3 PHASE_B · M3 PHASE_C	Motor A · B · C	bare_wire_lead	CONFIGURED
Motor 4	4-in-1 ESC M4 PHASE_A · M4 PHASE_B · M4 PHASE_C	Motor A · B · C	bare_wire_lead	CONFIGURED

Step 3 of 11

TIP Any two leads of a motor may be swapped to reverse it. Set each motor to the spin given in the propeller step, in the ESC configurator rather than by re-soldering.

4 SOLDER THE BATTERY LEAD AND CAPACITOR

links: battery_pos, battery_neg, bulk_cap_pos, bulk_cap_neg

A
×1

Panasonic EEU-FR1V471 470 μ F 35 V low-ESR electrolytic capacitor

B
×1

XT60 battery lead (12 AWG)

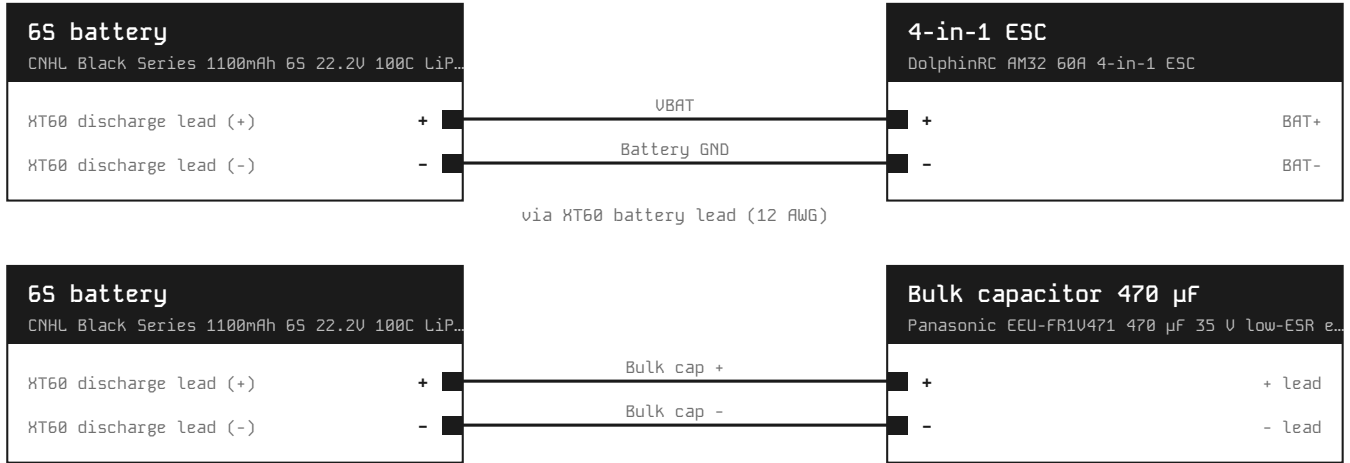


Figure 4 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

- Using the XT60 battery lead (12 AWG), connect VBAT, Battery GND as in the table (2 connections).
- Connect Bulk cap +, Bulk cap - as in the table (2 connections).

LINK	FROM	TO	TERMINATION	DRC
VBAT	6S battery +	4-in-1 ESC +	xt60 / solder_pad	CONFIGURED
Battery GND	6S battery -	4-in-1 ESC -	xt60 / solder_pad	CONFIGURED
Bulk cap +	6S battery +	Bulk capacitor 470 μ F +	xt60 / radial_lead	CONFIGURED
Bulk cap -	6S battery -	Bulk capacitor 470 μ F -	xt60 / radial_lead	CONFIGURED

Step 4 of 11

CAUTION Observe polarity on the capacitor and the XT60 lead before the first power-up.

NOTE Included with the stack (product page: '12AWG 12cm red/black lead and XT60 male plug'). Solder the bundled 470 μ F capacitor across the same pads. – dolphinrc-xt60-battery-lead

5 INSTALL THE AIR UNIT AND CAMERA

notes: camera_mount, vtx_mount

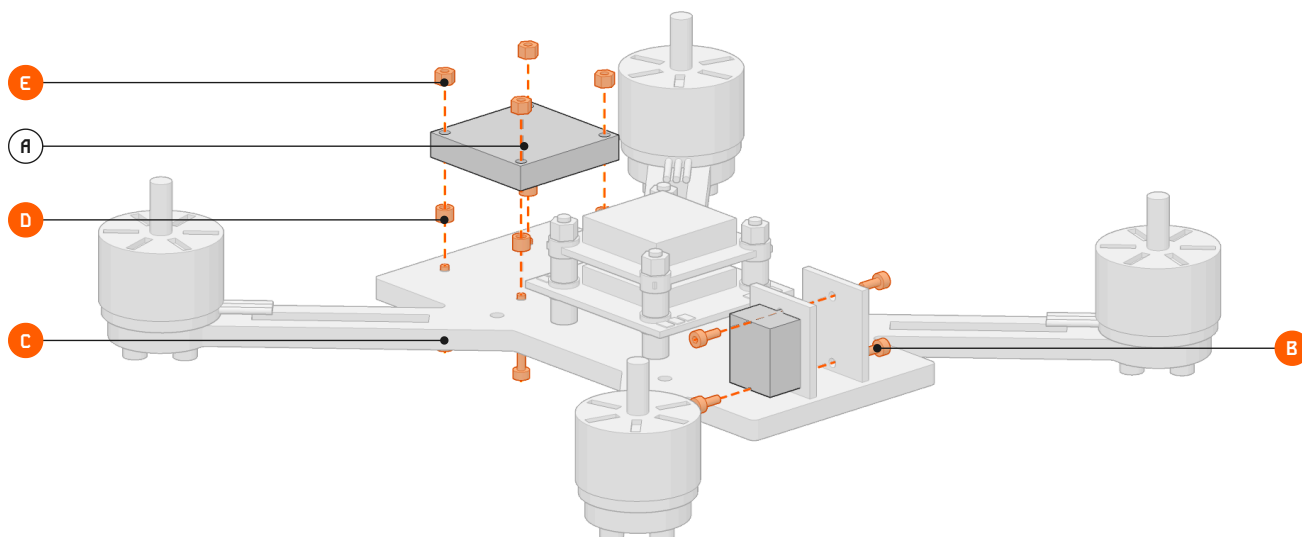
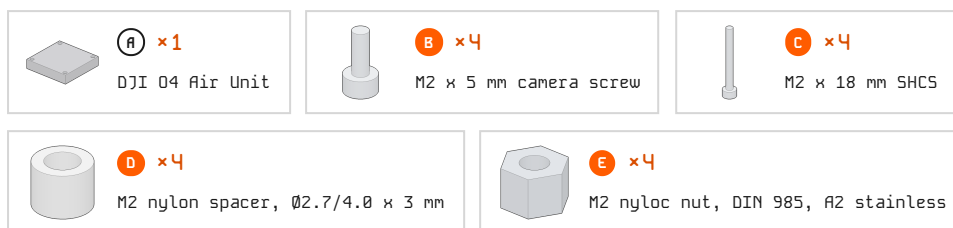


Figure 5 Install the air unit and camera: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Insert 2 × M2 × 5 mm camera screw through the camera side plates from one side.
2. Seat the DJI O4 Air Unit onto the camera side plates.
3. Insert 2 × M2 × 5 mm camera screw from the opposite side.
4. Insert 4 × M2 × 18 mm SHCS from below, up through the transmission module mount.
5. Slide 4 × M2 nylon spacer, Ø2.7/4.0 × 3 mm over the screws.
6. Lower the DJI O4 Air Unit onto the transmission module mount (3 mm gap).
7. Finish with 4 × M2 nyloc nut, DIN 985, A2 stainless; tighten until snug.

Torque 0.1 N·m ^A M2 × 5 mm camera screw · Camera hardware (M2); 0.1 N·m ^A M2 nyloc nut, DIN 985, A2 stainless · Air unit hardware (M2)

Tools 1.5 mm hex key, 4 mm nut driver **Step** 5 of 11



NOTE DJI O4 manual: 'DO NOT overtighten the screws to avoid stripping.' No threadlocker: the camera's thread material is not stated. – quadcopter-5in-camera-hardware



CAUTION Needs airflow. Install inside the frame in a well-ventilated spot (near air intake/exhaust), within ~1 cm of the propellers to use downwash, and not in an enclosed area; a heat-conductive pad to the carbon frame extends run time. Ground standby from cold start about 2.5 min at 25 °C (ground low-power mode on); above 25 °C ambient use external cooling. – dji-o4-air-unit

6

CONNECT THE AIR UNIT

links: video_power, video_gnd, video_osd

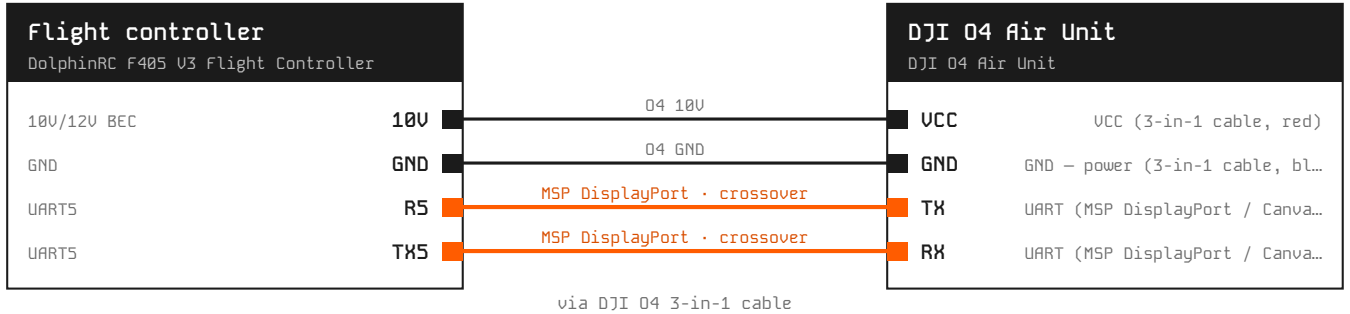


Figure 6 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Using the DJI O4 3-in-1 cable, connect O4 10V, O4 GND, MSP DisplayPort as in the table (4 connections).

LINK	FROM	TO	TERMINATION	DRC
O4 10V	Flight controller 10V	DJI O4 Air Unit VCC	solder_pad	CONFIGURED
O4 GND	Flight controller GND	DJI O4 Air Unit GND	solder_pad	CONFIGURED
MSP DisplayPort	Flight controller R5 · TX5	DJI O4 Air Unit TX · RX	* crossover dji_6pin / dji_3in1_cable_6pin	CONFIGURED

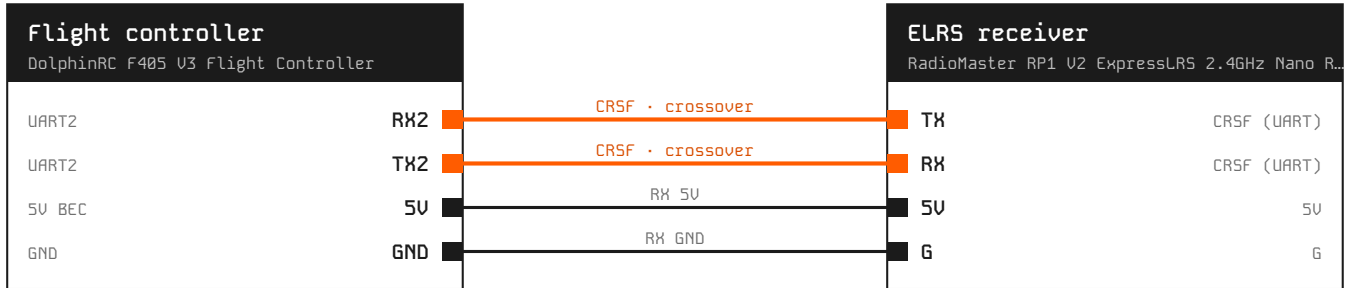
Step 6 of 11

NOTE Ships with the DJI O4 Air Unit. Wire order VCC, GND, RX, TX, GND, S.Bus (O4 user manual). – dji-o4-3in1-cable

7

CONNECT THE RECEIVER

links: rx_crsf, rx_power, rx_gnd



LINK	FROM	TO	TERMINATION	DRC
CRSF	Flight controller RX2 · TX2	ELRS receiver TX · RX	* CROSSOVER uart	CONFIGURED
RX 5V	Flight controller 5V	ELRS receiver 5V	solder_pad	CONFIGURED
RX GND	Flight controller GND	ELRS receiver G	solder_pad	CONFIGURED

Step 7 of 11

NOTE Mounting: none (no holes); heat shrink / foam tape / zip tie. – radiomaster-rp1-v2-elrs-2g4

8 FIT THE TOP PLATE

mates: top_plate_mount

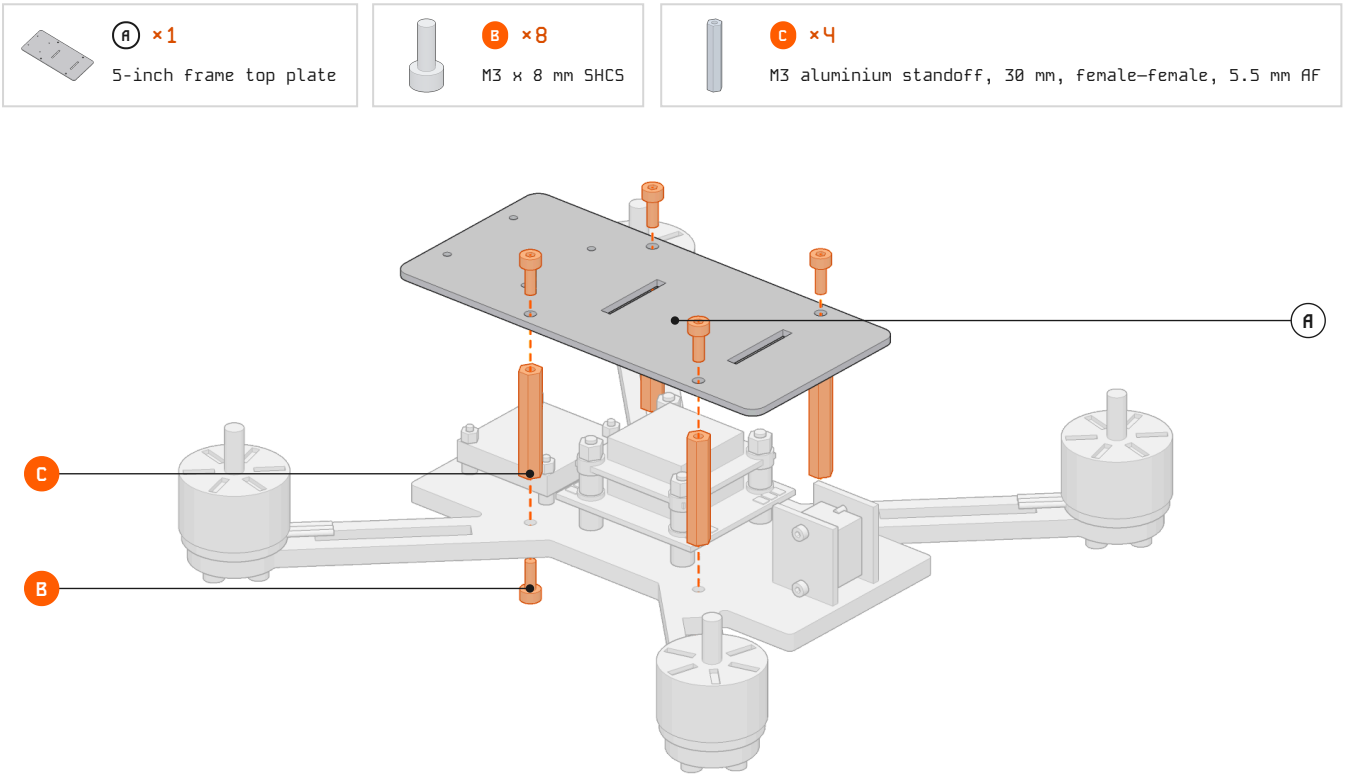


Figure 8 Fit the top plate: parts explthd along each joint's axis in stacking order; lettered as in the parts strip.

1. Insert 4 × M3 × 8 mm SHCS from below, up through the top-plate standoffs.
2. Thread 4 × M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF onto the screws.
3. Lower the 5-inch frame top plate onto the top-plate standoffs (30 mm gap).
4. Drive 4 × M3 × 8 mm SHCS from above, head down.

Torque 0.6 N·m ^A M3 × 8 mm SHCS Tools 2.5 mm hex key, 5.5 mm spanner Step 8 of 11

TIP Threadlocker: medium strength (blue) on the standoff screws (lower and upper). Steel into aluminium with no locking element, so vibration can back the screws out. Advice is from X-Team's motor-screw guide, a secondary source; no maker of these parts states it. – quadcopter-5in-frame-hardware

9 MOUNT AND CONNECT THE GNSS

mates: gps_mount · links: gnss_uart, gnss_i2c, gnss_power, gnss_gnd

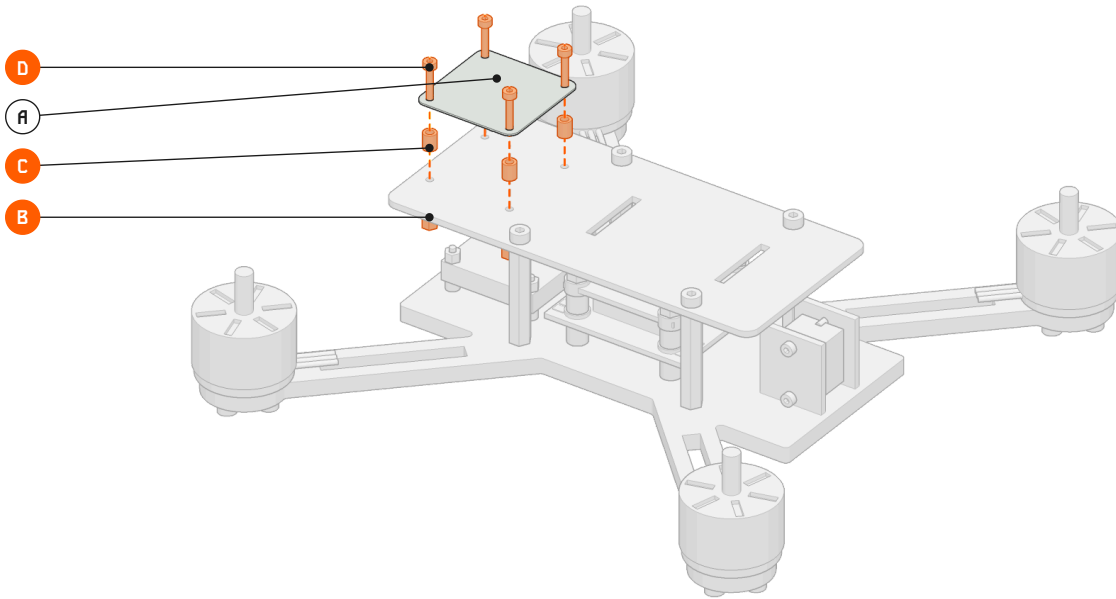
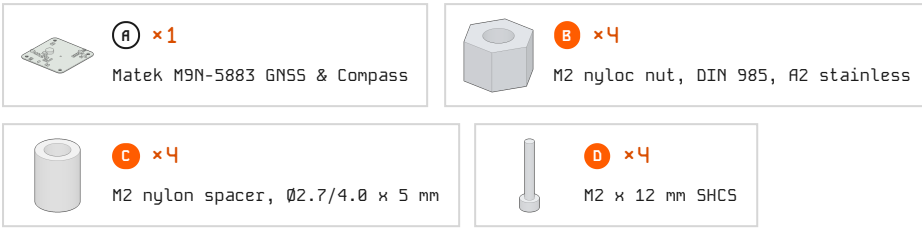


Figure 9 Mount and connect the GNSS: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Hold 4 × M2 nyloc nut, DIN 985, A2 stainless under the GPS mount.
2. Fit 4 × M2 nylon spacer, Ø2.7/4.0 x 5 mm on the GPS mount.
3. Lower the Matek M9N-5883 GNSS & Compass onto the GPS mount (5 mm gap).
4. Drive 4 × M2 x 12 mm SHCS from above, head down.
5. Connect GPS, Compass, GPS 4.5V, GPS GND as in the table (6 connections).

LINK	FROM	TO	TERMINATION	DRC
GPS	Flight controller RX1 · TX1	GNSS + compass TX · RX	* crossover uart	CONFIGURED
Compass	Flight controller SDA · SCL	GNSS + compass DA · CL	i2c	CONFIGURED
GPS 4.5V	Flight controller 4.5V	GNSS + compass 5V	solder_pad / jst_gh_6	CONFIGURED
GPS GND	Flight controller GND	GNSS + compass G	solder_pad / jst_gh_6	CONFIGURED

Torque 0.1 N·m ^A M2 x 12 mm SHCS Tools 1.5 mm hex key, 4 mm nut driver Step 9 of 11

10 STRAP ON THE BATTERY

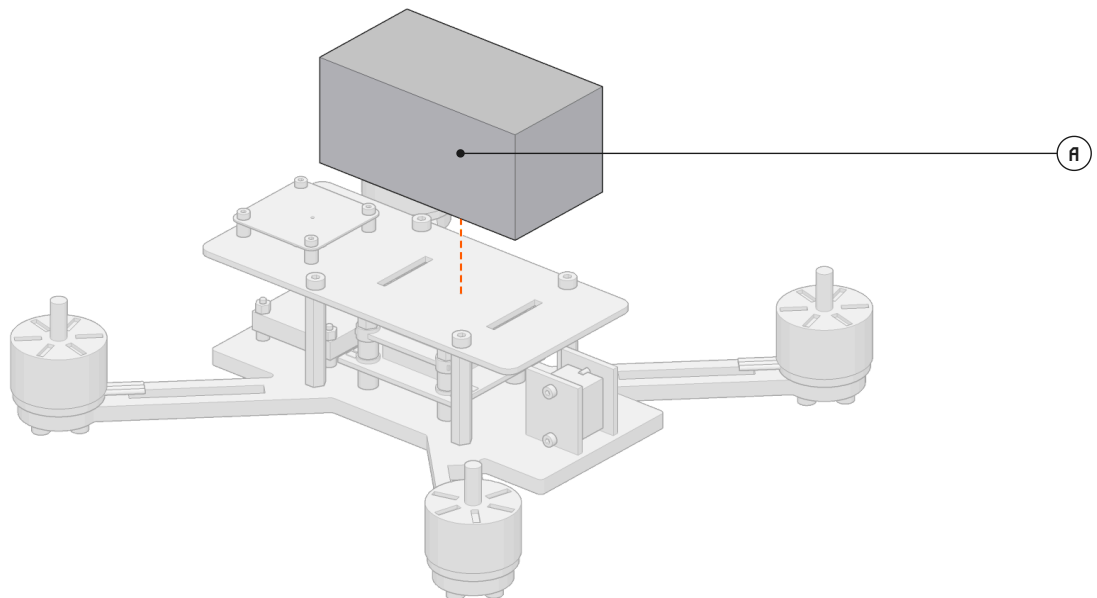
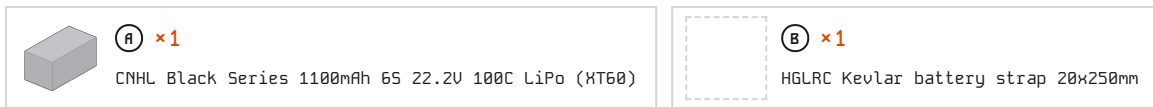


Figure 10 Strap on the battery: shown lifted above its seat.

1. Place the CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60) as shown. It has no mechanical interface in the model; the figure places it by frame BATTERY_PAD (strapped, no mechanical interface).
2. Fit the HGLRC Kevlar battery strap 20x250mm.

Step 10 of 11

CAUTION Fit and connect the battery only for flight; remove the props for any bench work.

NOTE Mounting: none – strapped around the body (no strap interface vocabulary). – cnhl-black-series-1100mah-6s-100c

11 FIT THE PROPELLERS

mates: arm_fr/prop_on_shaft,
arm_rr/prop_on_shaft, arm_rl/prop_on_shaft,
arm_fl/prop_on_shaft

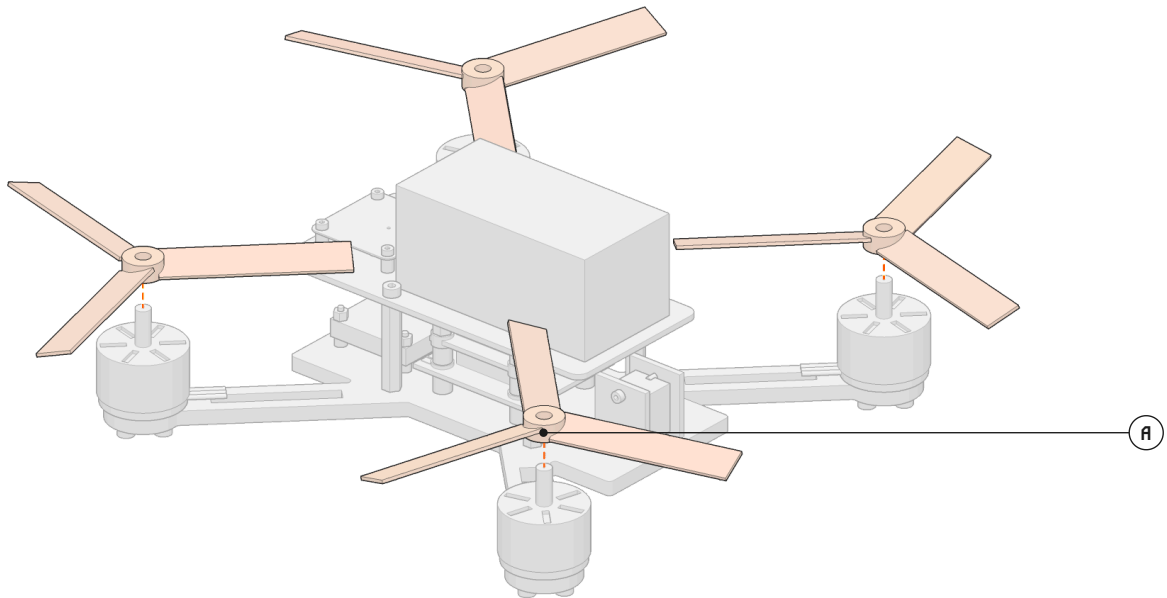
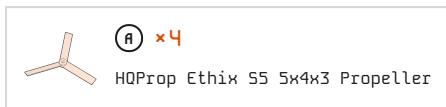


Figure 11 Fit the propellers: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Seat a HQProp Ethix S5 5x4x3 Propeller onto each output shaft Ø5 mm M5. Repeat on all 4.
2. Front-right arm (M2): motor spins CCW from above, so fit a **CCW** HQProp Ethix S5 5x4x3 Propeller.
3. Rear-right arm (M1): motor spins CW from above, so fit a **CW** HQProp Ethix S5 5x4x3 Propeller.
4. Rear-left arm (M3): motor spins CCW from above, so fit a **CCW** HQProp Ethix S5 5x4x3 Propeller.
5. Front-left arm (M4): motor spins CW from above, so fit a **CW** HQProp Ethix S5 5x4x3 Propeller.

Step 11 of 11



WARNING Fit propellers last, after configuration and a motor-direction check with the props off.



CAUTION A pack holds 2 CW and 2 CCW props; the steps above say which goes on each motor. The directions are Betaflight's Quad-X default, props in (generated/betaflight-cli.txt sets yaw_motors_reversed to match). Check each motor's direction with the props off before fitting them.

15 COVERAGE

ASSEMBLE() · WIRINGCHECKLIST

Every rigid mate, every hardware placement and every electrical link in the model, and the step that covers it. The verifier fails the guide if any row is not covered.

MATES

MATE	INTERFACES	HARNESS	STEP
arm_fr_mount	arm_fr/motor.base_mount ↔ frame.motor_mount_fr	arm_fr_hardware	1 · Mount the motors
arm_rr_mount	arm_rr/motor.base_mount ↔ frame.motor_mount_rr	arm_rr_hardware	1 · Mount the motors
arm_rl_mount	arm_rl/motor.base_mount ↔ frame.motor_mount_rl	arm_rl_hardware	1 · Mount the motors
arm_fl_mount	arm_fl/motor.base_mount ↔ frame.motor_mount_fl	arm_fl_hardware	1 · Mount the motors
stack_mount	stack/esc.stack_mount ↔ frame.stack_mount	stack_hardware	2 · Build the FC and ESC stack
fc_mount	stack/fc.stack_mount ↔ frame.stack_mount	stack_hardware	2 · Build the FC and ESC stack
top_plate_mount	frame.standoff_mount ↔ top_plate.standoff_mount	frame_hardware	8 · Fit the top plate
camera_mount	video.camera_mount ↔ frame.camera_mount	camera_hardware	5 · Install the air unit and camera
vtx_mount	video.tx_module_mount ↔ frame.vtx_mount	vtx_hardware	5 · Install the air unit and camera
gps_mount	gnss.mount ↔ top_plate.gps_mount	gps_hardware	9 · Mount and connect the GNSS
arm_fr/prop_on_shaft	arm_fr/motor.shaft ↔ arm_fr/prop.hub_bore	—	11 · Fit the propellers
arm_rr/prop_on_shaft	arm_rr/motor.shaft ↔ arm_rr/prop.hub_bore	—	11 · Fit the propellers
arm_rl/prop_on_shaft	arm_rl/motor.shaft ↔ arm_rl/prop.hub_bore	—	11 · Fit the propellers
arm_fl/prop_on_shaft	arm_fl/motor.shaft ↔ arm_fl/prop.hub_bore	—	11 · Fit the propellers

HARDWARE PLACEMENTS

HARNESS / CHILD	PARTS	JOINT	STEP
arm_fr_hardware/screws	4 × M3 × 8 mm motor screw	arm_fr_mount	1 · Mount the motors
arm_rr_hardware/screws	4 × M3 × 8 mm motor screw	arm_rr_mount	1 · Mount the motors
arm_rl_hardware/screws	4 × M3 × 8 mm motor screw	arm_rl_mount	1 · Mount the motors
arm_fl_hardware/screws	4 × M3 × 8 mm motor screw	arm_fl_mount	1 · Mount the motors
stack_hardware/screws	4 × M3 × 30 mm SHCS	stack_mount	2 · Build the FC and ESC stack
stack_hardware/spacers	8 × ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25	stack_mount	2 · Build the FC and ESC stack
stack_hardware/nuts	4 × M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	stack_mount	2 · Build the FC and ESC stack
frame_hardware/screws	8 × M3 × 8 mm SHCS	top_plate_mount	8 · Fit the top plate
frame_hardware/standoffs	4 × M3 aluminium standoff, 30 mm, female-female, 5.5 mm AF	top_plate_mount	8 · Fit the top plate
camera_hardware/screws	4 × M2 × 5 mm camera screw	camera_mount	5 · Install the air unit and camera
vtx_hardware/screws	4 × M2 × 18 mm SHCS	vtx_mount	5 · Install the air unit and camera
vtx_hardware/spacers	4 × M2 nylon spacer, Ø2.7/4.0 × 3 mm	vtx_mount	5 · Install the air unit and camera
vtx_hardware/nuts	4 × M2 nyloc nut, DIN 985, A2 stainless	vtx_mount	5 · Install the air unit and camera

HARDWARE (CONTINUED)

HARNESS / CHILD	PARTS	JOINT	STEP
gps_hardware/nuts	4 × M2 nyloc nut, DIN 985, A2 stainless	gps_mount	9 · Mount and connect the GNSS
gps_hardware/spacers	4 × M2 nylon spacer, Ø2.7/4.0 × 5 mm	gps_mount	9 · Mount and connect the GNSS
gps_hardware/screws	4 × M2 × 12 mm SHCS	gps_mount	9 · Mount and connect the GNSS

WIRING LINKS

LINK	NAME	HARNESS	STEP
battery_pos	VBAT	xt60_lead	4 · Solder the battery lead and capacitor
battery_neg	Battery GND	xt60_lead	4 · Solder the battery lead and capacitor
bulk_cap_pos	Bulk cap +	—	4 · Solder the battery lead and capacitor
bulk_cap_neg	Bulk cap -	—	4 · Solder the battery lead and capacitor
m2_phases	Motor 2	—	3 · Solder the motor leads
m1_phases	Motor 1	—	3 · Solder the motor leads
m3_phases	Motor 3	—	3 · Solder the motor leads
m4_phases	Motor 4	—	3 · Solder the motor leads
rx_crsf	CRSF	—	7 · Connect the receiver
rx_power	RX 5V	—	7 · Connect the receiver
rx_gnd	RX GND	—	7 · Connect the receiver
video_power	04 10V	dji_cable	6 · Connect the air unit
video_gnd	04 GND	dji_cable	6 · Connect the air unit
video_osd	MSP DisplayPort	dji_cable	6 · Connect the air unit
gnss_uart	GPS	—	9 · Mount and connect the GNSS
gnss_i2c	Compass	—	9 · Mount and connect the GNSS
gnss_power	GPS 4.5V	—	9 · Mount and connect the GNSS
gnss_gnd	GPS GND	—	9 · Mount and connect the GNSS
stack/fc_esc_cable	FC/ESC 8-pin cable	stack/sh8_cable	2 · Build the FC and ESC stack